

# FOX\_Checklist

**What this is:** every FoxByte feature and its sub-features — built or not, and how it was tested — so it's clear where we are and what's next. How each feature works is in `FOX_Feature_Implemented.pdf` (same ids).

**Last updated:** 22 September 2026 · **Reflects:** `main @ 2778762` (v1.0.1, exact repository path PR #1) + the FoxByte rename and brand indirection ( `chore/foxbyte` )

**Source:** `docs/FOX_Checklist.html` → `make feature-doc` → this PDF

✓

Done & tested — automated tests (unit, VM integration or CI)

🔸

Partial — gaps, or not verified end to end

❌

Known issue

☐

Not started

Sub-features:    ✓    built    🔸    partly    ☐    not built yet    ❌    broken    · "VM §n" = section of    `make integration`    ; "v2 §n" = `make integration-v2`    ; "update" = `make integration-update`

## Where we are

| Area                                    | Features | Done & tested | Done (manual) | Partial | Known issue | Not started | Progress                    |
|-----------------------------------------|----------|---------------|---------------|---------|-------------|-------------|-----------------------------|
| A · Install & setup                     | 9        | 4             | 3             | 2       | ·           | ·           | <div><div></div></div> 78%  |
| B · CLI & services                      | 2        | 2             | ·             | ·       | ·           | ·           | <div><div></div></div> 100% |
| C · Branching & storage                 | 8        | 7             | ·             | 1       | ·           | ·           | <div><div></div></div> 88%  |
| D · Durability                          | 6        | 5             | ·             | 1       | ·           | ·           | <div><div></div></div> 83%  |
| E · Security foundations                | 3        | 3             | ·             | ·       | ·           | ·           | <div><div></div></div> 100% |
| F · SQL gateway                         | 7        | 6             | 1             | ·       | ·           | ·           | <div><div></div></div> 100% |
| G · Accounts & auth                     | 5        | 4             | ·             | 1       | ·           | ·           | <div><div></div></div> 80%  |
| H · Blackbox (record of schema changes) | 18       | 17            | ·             | 1       | ·           | ·           | <div><div></div></div> 94%  |
| I · Import & migration                  | 13       | 5             | 8             | ·       | ·           | ·           | <div><div></div></div> 100% |
| J · ETL pipelines (experimental)        | 4        | 2             | 2             | ·       | ·           | ·           | <div><div></div></div> 100% |
| K · APIs, SDKs & agents                 | 6        | 4             | 2             | ·       | ·           | ·           | <div><div></div></div> 100% |
| L · Web console                         | 11       | 1             | 10            | ·       | ·           | ·           | <div><div></div></div> 100% |
| M · Build, CI & releases                | 7        | 6             | 1             | ·       | ·           | ·           | <div><div></div></div> 100% |
| Total (features built or started)       | 99       | 66            | 27            | 6       | ·           | ·           | <div><div></div></div> 94%  |
| N · Not started (roadmap)               | 11       | ·             | ·             | ·       | ·           | 11          | <div><div></div></div> 0%   |

Storage engine — not implemented, by design

FoxByte does **not** have its own storage engine. Data is stored by **PostgreSQL's own engine** (heap tables, B-tree indexes, write-ahead log) running on local disk inside the stock `postgres:18-bookworm` image plus `wal-g` ( `postgres:16-bookworm` for an install created before 21 Sep 2026). This was a locked architecture decision: a Neon-style engine that separates compute from storage over the network adds latency to every transaction, and transaction speed must never be compromised.  
  
What **is** built is a storage *layer* around Postgres: copy-on-write branch drivers — ZFS on macOS/Linux, btrfs on Windows — behind one interface in `internal/branch/storage.go` (C1), quotas (C6), capacity reporting (C7), and durability to object storage with `wal-g` + MinIO: WAL archiving, base backups, point-in-time restore (D1–D3). Networked storage disaggregation is listed in the README roadmap but not started (see "Not started").

## Suggested next steps

1. **Security hardening from the 22 Sep audit** — safe defaults and per-user access are done; next, in order: Blackbox coverage of `TRUNCATE` /DML and signed off-host anchors (G04, G17, G28); backups to a remote target on a schedule, with one taken at install (G18, G19); signed releases, digest-pinned images and the VM suites in CI (G22–G24).

2. Run on real machines what the throwaway VM cannot: `fox update` across a real release on macOS and Windows (A9); on Windows, the `zstd` distro image's unpack (M5), the `btrfs` holding area of `fox pg upgrade` (D6), and an export's copy to the host (D5).
3. Make the `postgres-wal-g` image package public, so installs pull it instead of building it for a few minutes (C8).
4. Run the VM integration suites in CI on every pull request (M2).
5. Add automated tests where there are none: imports (I1–I3, I7), SDKs (K3, K4), web UI (L\*).
6. HA beyond the demo: automatic failover and fencing (D4).

## A Install & setup

7 of 9 done

| ID                                                                                            | Feature · sub-features                                                                                                                                                                                                                                                                                                                                          | Tested by                                                                            |
|-----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
|  <b>A1</b>   | <b>One-line installer (macOS / Linux)</b> <ul style="list-style-type: none"> <li>✓ Detects OS / CPU, downloads the release binary</li> <li>✓ Installs to the prefix (sudo only if needed)</li> <li>✓ Stages the Linux engine for the VM on macOS</li> </ul>                                                                                                     | Unit ( <code>test_install_sh.sh</code> , CI); manual                                 |
|  <b>A2</b>   | <b>Windows installer (install.ps1)</b> <ul style="list-style-type: none"> <li>✓ Installs WSL, elevates, resumes after reboot</li> <li>✓ Downloads fox.exe, engine, image build context</li> <li>✓ Ubuntu rootfs verified against Ubuntu's checksums</li> <li>✓ Adds to PATH and runs <code>fox setup</code></li> </ul>                                          | Pester (Windows CI); Windows run 16 Sep                                              |
|  <b>A3</b>   | <b>fox setup on macOS (Lima)</b> <ul style="list-style-type: none"> <li>✓ Creates / starts the Lima VM</li> <li>✓ Installs Docker and ZFS</li> <li>✓ Reinstalls the engine on every run</li> <li>✓ Starts the stack</li> </ul>                                                                                                                                  | Unit; run against a deliberately broken VM, 21 Sep                                   |
|  <b>A4</b>   | <b>fox setup on Windows (WSL2 + btrfs)</b> <ul style="list-style-type: none"> <li>✓ Imports a dedicated distro with systemd</li> <li>✓ Docker, btrfs and a storage mount unit</li> <li>✓ Quiet steps, detail in install.log</li> <li>✓ Waits for systemd on a cold distro</li> </ul>                                                                            | Unit ( <code>host_wsl_test.go</code> ); Windows run 16 Sep                           |
|  <b>A5</b> | <b>Linux native</b> <ul style="list-style-type: none"> <li>✓ Engine runs in-process, no VM</li> <li>✓ <code>fox setup</code> explains nothing is needed</li> </ul>                                                                                                                                                                                              | VM integration (all suites run on Linux)                                             |
|  <b>A6</b> | <b>Command forwarding &amp; host file streaming</b> <ul style="list-style-type: none"> <li>✓ macOS → Lima VM, starts the VM if stopped</li> <li>✓ Windows → WSL distro</li> </ul>                                                                                                                                                                               | Unit ( <code>TestImportLocalFile</code> , <code>TestWSLargs</code> )                 |
|  <b>A7</b> | <b>Always-latest engine binary</b> <ul style="list-style-type: none"> <li>✓ Downloads the newest engine into the cache (ELF check)</li> <li>✓ <code>FOX_NO_REFRESH</code> , <code>FOX_VERSION</code></li> </ul>                                                                                                                                                 | Unit ( <code>host_test.go</code> , <code>update_test.go</code> ); Windows run 16 Sep |
|  <b>A8</b> | <b>New-release notice on fox start</b> <ul style="list-style-type: none"> <li>✓ Background check, 1.5 s limit, printed last</li> <li>✓ Only a promoted, newer, complete release</li> </ul>                                                                                                                                                                      | Unit; update \$1, \$1b, \$2, \$3, \$8; real Mac check                                |
|  <b>A9</b> | <b>fox update</b> <ul style="list-style-type: none"> <li>✓ Downloads and verifies everything before changing anything</li> <li>✓ Stops servers, installs engine, restarts old one on failure</li> <li>✓ <code>up</code> , Blackbox upgrade, <code>start</code> on the new engine</li> <li>✓ Replaces host <code>fox</code> , keeps previous binaries</li> </ul> | Unit; update \$4–\$7; a real Mac run                                                 |

## B CLI & services

2 of 2 done

| ID | Feature · sub-features | Tested by |
|----|------------------------|-----------|
|----|------------------------|-----------|

|   |                                        |                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                     |                                                                             |
|---|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| ✓ | <b>B1 The fox command set</b>          | <ul style="list-style-type: none"> <li>✓ Stack: setup, start, stop, up, down, status, logs, psql, update</li> <li>✓ Branching, backup / restore, HA</li> <li>✓ Blackbox, policy, impact</li> <li>✓ Import, pipelines</li> <li>✓ backup export / restore, pg status / upgrade</li> </ul> | <ul style="list-style-type: none"> <li>✓ Users, API keys, admin</li> <li>✓ <code>fox vm status / fox vm shell</code> (Lima, WSL; Linux says there is no VM)</li> <li>✓ <code>branch create --from</code></li> <li>✓ <code>fox uninstall</code> removes it all, retired installs included</li> </ul> | Unit ( <code>main_test.go</code> , <code>update_test.go</code> ); VM suites |
| ✓ | <b>B2 Background services (daemon)</b> | <ul style="list-style-type: none"> <li>✓ Control plane :8080, gateway :6432, agent API :8088</li> <li>✓ Detached, pidfiles and logs in <code>~/ .fox</code></li> </ul>                                                                                                                  | <ul style="list-style-type: none"> <li>✓ Stop (TERM then KILL), status, logs</li> <li>□ Health check / restart on crash</li> </ul>                                                                                                                                                                  | Unit ( <code>TestStopTerminatesProcess</code> ); VM §1                      |

## C Branching & storage

7 of 8 done

| ID   | Feature · sub-features                                                                                                                                                                                                                            | Tested by                                                                   |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| ✓ C1 | <b>Storage drivers (ZFS / btrfs)</b> <ul style="list-style-type: none"> <li>✓ ZFS snapshot + clone per branch (macOS VM, Linux)</li> <li>✓ btrfs subvolume snapshots (Windows)</li> <li>✓ Pool / filesystem created on a loopback file</li> </ul> | Unit; VM §2, §2b (ZFS); btrfs manual on Windows                             |
| ✓ C2 | <b>Branch lifecycle</b> <ul style="list-style-type: none"> <li>✓ Create, list, delete</li> <li>✓ Reset to parent</li> <li>✓ Suspend / resume</li> <li>✓ Recreate a container from its surviving data</li> </ul>                                   | Unit; VM §2, §2b, §4                                                        |
| ✓ C3 | <b>Protection of main</b> <ul style="list-style-type: none"> <li>✓ Delete and reset refused</li> </ul>                                                                                                                                            | Unit; VM §4, §6                                                             |
| ✓ C4 | <b>Private networking (no host ports)</b> <ul style="list-style-type: none"> <li>✓ Branch databases reachable only through the gateway</li> <li>✓ Published ports (object store, restore, debug) on loopback only; standby unpublished</li> </ul> | Unit ( <code>TestParsePublishedPort</code> , <code>TestDSN</code> ); VM §1c |
| ✓ C5 | <b>Scale-to-zero &amp; wake on connect</b> <ul style="list-style-type: none"> <li>✓ Idle branches suspended automatically</li> <li>✓ Next connection wakes the branch (~0.3–0.7 s)</li> </ul>                                                     | VM §7                                                                       |
| 🔥 C6 | <b>Storage quotas</b> <ul style="list-style-type: none"> <li>✓ Space reserved for main (ZFS)</li> <li>🔥 Per-branch cap (opt-in)</li> </ul>                                                                                                        | Manual                                                                      |
| ✓ C7 | <b>Capacity &amp; status</b> <ul style="list-style-type: none"> <li>✓ Services, main health, backups, branches</li> <li>✓ Storage used / available</li> </ul>                                                                                     | VM §1                                                                       |
| ✓ C8 | <b>Provisioning &amp; image pull</b> <ul style="list-style-type: none"> <li>✓ Docker, storage pool and image prepared on up</li> </ul>                                                                                                            | Unit; VM §11b                                                               |

## D Durability

5 of 6 done

| ID   | Feature · sub-features                                                                                                                                                  | Tested by      |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| ✓ D1 | <b>WAL archiving to object storage</b> <ul style="list-style-type: none"> <li>✓ wal-g ships main's WAL to MinIO, off the commit path</li> </ul>                         | VM §3          |
| ✓ D2 | <b>Base backups</b> <ul style="list-style-type: none"> <li>✓ Create and list</li> <li>✓ Follow the primary after a failover; listing works with main stopped</li> </ul> | VM §3, §3b, §6 |

|   |                                         |                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                          |                               |
|---|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| ✓ | <b>D3 Point-in-time restore</b>         | <ul style="list-style-type: none"> <li>✓ Restore to a timestamp or latest on :5433</li> <li>✓ Per-install credentials</li> <li>✓ Starts from the newest base backup that precedes the target</li> </ul>                                                                                                     | <ul style="list-style-type: none"> <li>✓ A point no backup precedes is refused, naming the oldest</li> </ul>                                                                                                                                                                                                                                                                             | Unit; VM §3, §3b, §6; v2 §1a  |
| 🔥 | <b>D4 HA standby &amp; failover</b>     | <ul style="list-style-type: none"> <li>✓ Streaming hot standby</li> <li>✓ Manual failover, <code>main</code> rerouted</li> <li>✓ Nothing deletes or stops the promoted standby</li> <li>✓ <code>fox up</code> brings it back, not the old main</li> </ul>                                                   | <ul style="list-style-type: none"> <li>✓ <code>fox ha failback</code>, keeping every write</li> <li>✓ The promoted standby archives WAL, so backups and PITR cover the failover</li> <li>❑ Automatic failover, fencing, multi-host</li> </ul>                                                                                                                                            | Unit; VM §6, §7               |
| ✓ | <b>D5 Portable export &amp; restore</b> | <ul style="list-style-type: none"> <li>✓ Every non-agent branch: <code>pg_dump</code>, its roles, the Blackbox head; SHA256SUMS over all</li> <li>✓ Copied out of the VM and verified on the host; a copy that fails is deleted</li> <li>✓ Damaged, incomplete or path-escaping archives refused</li> </ul> | <ul style="list-style-type: none"> <li>✓ Restore into a new branch, same or newer major; Blackbox checked exact</li> <li>🔥 Host copy on macOS/Windows unit-tested only</li> </ul>                                                                                                                                                                                                        | Unit; VM pg-upgrade §1        |
| ✓ | <b>D6 PostgreSQL major upgrade</b>      | <ul style="list-style-type: none"> <li>✓ <code>fox pg status</code>: the major of the install and of each branch</li> <li>✓ Plan with every change crossed, found-in-your-data first; refusals say what to do</li> <li>✓ Asks, exports, then reloads every carried branch on the new major</li> </ul>       | <ul style="list-style-type: none"> <li>✓ Data, roles, API keys and every Blackbox entry come across</li> <li>✓ Old databases held outside the branch namespace; rollback and finalize</li> <li>✓ Each upgraded cluster archives WAL under its own prefix</li> <li>🔥 <code>btrfs</code> (Windows) holding area not yet run</li> <li>❑ Carry agent branches; upgrade with HA on</li> </ul> | Unit; VM pg-upgrade (72 / 72) |

## E Security foundations 3 of 3 done

| ID | Feature · sub-features                                                                                                                                                                                                                                     | Tested by                                                                                                                                                                                             |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ✓  | <b>E1 Per-install generated secrets</b>                                                                                                                                                                                                                    | Unit (indirect); v2 §1a; VM §1d                                                                                                                                                                       |
|    | <ul style="list-style-type: none"> <li>✓ Generated Postgres and MinIO passwords</li> <li>✓ Used by restore too</li> <li>✓ Each login role has its own derived password</li> </ul>                                                                          | <ul style="list-style-type: none"> <li>✓ Never on a command line (docker <code>--env-file</code>)</li> </ul>                                                                                          |
| ✓  | <b>E2 TLS everywhere</b>                                                                                                                                                                                                                                   | Unit<br>( <code>TestReadStartupUpgradesTLS</code> )                                                                                                                                                   |
|    | <ul style="list-style-type: none"> <li>✓ Self-signed TLS on all three services, zero setup</li> </ul>                                                                                                                                                      | <ul style="list-style-type: none"> <li>✓ TLS required from other machines (loopback clients may skip it)</li> <li>✓ No plain-HTTP fallback except on loopback</li> </ul>                              |
| ✓  | <b>E3 Keys are made by a person, not by start</b>                                                                                                                                                                                                          | Integration §1                                                                                                                                                                                        |
|    | <ul style="list-style-type: none"> <li>✓ Start creates no account and no key, and caches none on disk</li> <li>✓ The banner shows the connection string and how to make a key</li> <li>✓ The first account created gets the admin grant on main</li> </ul> | <ul style="list-style-type: none"> <li>✓ First account over HTTP needs the setup token ( <code>fox setup-token</code> )</li> <li>✓ Exactly one first account, however many sign up at once</li> </ul> |

## F SQL gateway 7 of 7 done

| ID | Feature · sub-features                                                                                                                               | Tested by                                                                                                                                                           |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ✓  | <b>F1 Routing by database name</b>                                                                                                                   | Unit; VM §1, §6                                                                                                                                                     |
|    | <ul style="list-style-type: none"> <li>✓ One endpoint :6432, database name = branch</li> </ul>                                                       | <ul style="list-style-type: none"> <li>❑ Connection pooling, rate limiting</li> </ul>                                                                               |
| ✓  | <b>F2 API-key authentication</b>                                                                                                                     | VM §1                                                                                                                                                               |
|    | <ul style="list-style-type: none"> <li>✓ <code>key_</code> key as the Postgres password</li> <li>✓ Bad key rejected with a Postgres error</li> </ul> | <ul style="list-style-type: none"> <li>✓ Startup deadline and connection cap</li> <li>✓ An account key opens main, its own branches, or any for an admin</li> </ul> |
| ✓  | <b>F3 Backend login (SCRAM)</b>                                                                                                                      | VM §1, §6, §8 (exercised)                                                                                                                                           |
|    | <ul style="list-style-type: none"> <li>✓ Cleartext, md5, SCRAM-SHA-256</li> <li>✓ Clients never see the database password</li> </ul>                 | <ul style="list-style-type: none"> <li>❑ Unit test for SCRAM</li> </ul>                                                                                             |

|   |                                             |                                                                                                                                                         |                                                                                                 |                                     |
|---|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|-------------------------------------|
| ✓ | <b>F4 Per-user Postgres login roles</b>     | <ul style="list-style-type: none"> <li>✓ Role per user, actor can't be forged</li> <li>✓ Permissions of non-superuser <code>db_client</code></li> </ul> | <ul style="list-style-type: none"> <li>✓ Own password per role</li> </ul>                       | Manual                              |
| ✓ | <b>F5 Attribution session settings</b>      | <ul style="list-style-type: none"> <li>✓ Actor kind, branch, session id per connection</li> </ul>                                                       | <ul style="list-style-type: none"> <li>○ Kind / branch / session client-settable</li> </ul>     | VM §8                               |
| ✓ | <b>F6 Ctrl-C (CancelRequest) forwarding</b> | <ul style="list-style-type: none"> <li>✓ Cancel a running query through the gateway</li> </ul>                                                          |                                                                                                 | Unit                                |
| ✓ | <b>F7 Errors &amp; insecure build tag</b>   | <ul style="list-style-type: none"> <li>✓ Real Postgres error responses</li> </ul>                                                                       | <ul style="list-style-type: none"> <li>✓ No-auth bypass impossible in release builds</li> </ul> | Unit ( <code>TestSendError</code> ) |

## G Accounts & auth 4 of 5 done

| ID                                            | Feature · sub-features                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Tested by                                                                                   |
|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| ✓ <b>G1 Auth store (SQLite)</b>               | <ul style="list-style-type: none"> <li>✓ Users, keys, sessions, OAuth ids, pipelines</li> <li>✓ Safe under concurrent use (WAL, busy timeout), including a new store created by three services at once</li> <li>□ Roles / permissions, multi-tenant projects</li> </ul>                                                                                                                                                                                                                                    | Unit ( <code>TestVerifyKeyConcurrent</code> , <code>TestConcurrentFirstOpen</code> ); VM §1 |
| ✓ <b>G2 Email &amp; password users</b>        | <ul style="list-style-type: none"> <li>✓ Create account, log in</li> <li>✓ Password change (signs out other sessions); admin reset at the CLI</li> <li>□ Self-service reset by email</li> <li>✓ List and delete users (admins)</li> <li>✓ Emails validated; at most 63 bytes (a role name)</li> </ul>                                                                                                                                                                                                      | Unit; VM §1; VM §1d                                                                         |
| ✓ <b>G3 API keys</b>                          | <ul style="list-style-type: none"> <li>✓ Create (shown once), list, revoke</li> <li>✓ Bearer / X-API-Key and gateway password</li> <li>○ Revoking an unknown id reports success</li> </ul>                                                                                                                                                                                                                                                                                                                 | Unit; VM §1                                                                                 |
| ✓ <b>G4 Web sessions &amp; signup control</b> | <ul style="list-style-type: none"> <li>✓ httpOnly session cookie</li> <li>✓ Sign-up closed by default ( <code>FOX_SIGNUP=open</code> )</li> <li>✓ Failed sign-ins throttled</li> </ul>                                                                                                                                                                                                                                                                                                                     | Unit ( <code>TestSessions</code> )                                                          |
| ○ <b>G5 OAuth (GitHub / Google)</b>           | <ul style="list-style-type: none"> <li>✓ Sign-in flows built</li> <li>✓ Defaults match what the engine serves ( <code>https://localhost:8080</code> ; web origin = public URL)</li> <li>✓ Session cookie <code>Secure</code> / <code>SameSite</code> follow the origin</li> <li>□ Tested with a real OAuth app</li> <li>✓ Only a provider-verified email joins or creates an account; closed sign-up respected</li> <li>✓ State cookie one-use, <code>Secure</code> , <code>SameSite=Lax</code></li> </ul> | Unit; VM §1b                                                                                |

## H Blackbox (record of schema changes) 17 of 18 done

| ID                                               | Feature · sub-features                                                                                                                                                                                                                                                                                | Tested by           |
|--------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| ✓ <b>H1 DDL recording</b>                        | <ul style="list-style-type: none"> <li>✓ Who, tool, branch, statement, status, risk</li> <li>✓ Inherited by every branch</li> <li>✓ A restart records nothing: the install script's own statements are not captured</li> <li>□ Role / database changes, some dropped object types</li> </ul>          | VM §8; v2 §9        |
| ○ <b>H2 Hash chain, append-only &amp; verify</b> | <ul style="list-style-type: none"> <li>✓ Rows can't be edited or deleted normally</li> <li>✓ <code>fox ledger verify</code></li> <li>○ Superuser rewrite caught only via anchors (H10–H12)</li> <li>✓ <code>bb.policy</code> changes recorded (append-only, not chained)</li> </ul>                   | VM §8; v2 §0, §6b   |
| ✓ <b>H3 Attribution</b>                          | <ul style="list-style-type: none"> <li>✓ Humans via their login role</li> <li>✓ Agents via their own <code>agent-&lt;id&gt;</code> role</li> <li>✓ Kind derived from the role, so neither can pose as the other</li> <li>✓ Tool and session are client-declared by design, and labelled so</li> </ul> | Unit; VM §8; v2 §1c |

|   |                                                                     |                                                                                                                                                                                                                                                                                                                         |                                       |
|---|---------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| ✓ | <b>H4 Destructive-DDL guardrail</b>                                 | <ul style="list-style-type: none"> <li>✓ DROP TABLE / DROP SCHEMA refused, attempt recorded</li> <li>✓ Override only superusers / <code>db_admin</code>, use recorded</li> <li>✓ "flag" and "allow" actions; unknown actions refused</li> <li>✓ No hang when blocked after a write in the same transaction</li> </ul>   | VM §8; v2 §1b, §6b                    |
| ✓ | <b>H5 Risk flags</b>                                                | <ul style="list-style-type: none"> <li>✓ Type change, drop column flagged</li> <li>✓ Judged on the running statement (no comments or literals)</li> <li>✓ Automated test</li> </ul>                                                                                                                                     | Unit; v2 §6b                          |
| ✓ | <b>H6 Blackbox access (CLI / REST / MCP / UI / SDK)</b>             | <ul style="list-style-type: none"> <li>✓ CLI, REST with filters, MCP, web page, SDKs</li> <li>□ Per-user access control</li> </ul>                                                                                                                                                                                      | v2 §0, §5                             |
| ✓ | <b>H7 <code>fox ledger revert --to</code></b>                       | <ul style="list-style-type: none"> <li>✓ Help and output say it's a point-in-time restore of main</li> <li>✓ Points to H13 for a branch from before a change</li> </ul>                                                                                                                                                 | Unit (help text); VM §3 (the restore) |
| ✓ | <b>H8 Override permission (<code>db_admin</code>) &amp; upgrade</b> | <ul style="list-style-type: none"> <li>✓ <code>fox admin grant revoke list</code></li> <li>✓ First user granted automatically</li> <li>✓ <code>fox blackbox upgrade [--all]</code></li> <li>✓ Grant / revoke from the web (admins only)</li> </ul>                                                                      | v2 §1b, §6                            |
| ✓ | <b>H9 Blackbox 2.0 capture</b>                                      | <ul style="list-style-type: none"> <li>✓ Transaction id and WAL position per entry</li> <li>✓ Task, tool-call hash, override use</li> <li>✓ Fail-safe, append-only</li> <li>✓ Kill switch not settable by clients</li> </ul>                                                                                            | Unit; v2 §2                           |
| ✓ | <b>H10 Checkpoints &amp; anchors</b>                                | <ul style="list-style-type: none"> <li>✓ Merkle root in the database and in read-only anchor files</li> <li>✓ On a schedule and on demand</li> <li>✓ Published anchor format</li> <li>□ Cloud write-once anchors (S3 Object Lock)</li> </ul>                                                                            | Unit; v2 §3, §3b                      |
| ✓ | <b>H11 <code>fox-verify</code> (independent verifier)</b>           | <ul style="list-style-type: none"> <li>✓ Checks a live branch or an export</li> <li>✓ Release binaries for every platform</li> </ul>                                                                                                                                                                                    | Unit; v2 §3                           |
| ✓ | <b>H12 Blackbox integrity &amp; export</b>                          | <ul style="list-style-type: none"> <li>✓ INTACT / TAMPERED via CLI, REST, MCP</li> <li>✓ JSON-lines export</li> <li>□ Streaming for very large ledgers</li> </ul>                                                                                                                                                       | v2 §3                                 |
| ✓ | <b>H13 Branch from just before a Blackbox entry</b>                 | <ul style="list-style-type: none"> <li>✓ Restore + replay up to the change's transaction</li> <li>✓ By transaction id or timestamp</li> <li>✓ CLI, REST, MCP, web panel</li> <li>✓ main untouched</li> </ul>                                                                                                            | Unit; v2 §4                           |
| ✓ | <b>H14 Blackbox names &amp; aliases</b>                             | <ul style="list-style-type: none"> <li>✓ <code>fox blackbox</code>, <code>/blackbox</code> routes, MCP and SDK names</li> <li>✓ Every original name still works</li> </ul>                                                                                                                                              | Unit; v2 §5                           |
| ✓ | <b>H15 Policy gate (warn / block)</b>                               | <ul style="list-style-type: none"> <li>✓ BBX02 warn / BBX01 block with JSON detail</li> <li>✓ Blocked attempts recorded</li> <li>✓ Per-rule <code>db_admin</code> override</li> <li>✓ CLI, REST, MCP, web page; custom rules</li> <li>✓ Patterns match the running statement</li> <li>□ TRUNCATE</li> </ul>             | Unit; v2 §6, §6b                      |
| ✓ | <b>H16 Agent provenance &amp; <code>execute_change</code></b>       | <ul style="list-style-type: none"> <li>✓ Session, task, parent session on every agent change (also through the Gateway DSN)</li> <li>✓ MCP <code>execute_change</code> with policy preview</li> <li>✓ <code>fox blackbox sessions</code>, REST</li> <li>🟡 Provenance is what the agent says (not verified)</li> </ul>   | Unit; v2 §7                           |
| ✓ | <b>H17 Impact analysis</b>                                          | <ul style="list-style-type: none"> <li>✓ Dependents (views, foreign keys, indexes, depth)</li> <li>✓ Other branches that have the object</li> <li>✓ Policy verdict and risk score</li> <li>✓ CLI, REST, MCP, inside <code>execute_change</code></li> <li>🟡 Functions, types, schemas not recognised from SQL</li> </ul> | Unit; v2 §8                           |
| ✓ | <b>H18 Blackbox diff between branches</b>                           | <ul style="list-style-type: none"> <li>✓ Fork point, changes only on each side</li> <li>✓ Objects changed on both</li> <li>✓ CLI, REST, MCP</li> <li>□ Merge (diff only today)</li> </ul>                                                                                                                               | Unit; v2 §8                           |

## I Import & migration

13 of 13 done

| ID                                      | Feature · sub-features                                                                                                                                                                                                                                                                                                                                              | Tested by            |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <input checked="" type="checkbox"/> I1  | <b>From Postgres</b> <ul style="list-style-type: none"> <li>✓ Schema and data via pg_dump</li> <li>✓ Zero-table result reported as an error</li> <li><input type="checkbox"/> Automated test</li> </ul>                                                                                                                                                             | Manual               |
| <input checked="" type="checkbox"/> I2  | <b>From MySQL ≤5.7 / MariaDB</b> <ul style="list-style-type: none"> <li>✓ pgloader with type mapping (multi-arch image)</li> <li><input type="checkbox"/> Automated test</li> </ul>                                                                                                                                                                                 | Manual               |
| <input checked="" type="checkbox"/> I3  | <b>From MySQL 8.x</b> <ul style="list-style-type: none"> <li>✓ Native mysqldump path (caching_sha2 auth)</li> <li>✓ Types, NULLs, unicode, PK</li> <li><input type="checkbox"/> Other indexes, foreign keys, defaults</li> <li><input type="checkbox"/> Automated test</li> </ul>                                                                                   | Manual               |
| <input checked="" type="checkbox"/> I4  | <b>From MongoDB</b> <ul style="list-style-type: none"> <li>✓ Collection → typed table, exact name</li> <li>✓ Nested objects / arrays kept as jsonb</li> </ul>                                                                                                                                                                                                       | VM §9, §10           |
| <input checked="" type="checkbox"/> I5  | <b>From a .sql dump</b> <ul style="list-style-type: none"> <li>✓ Runs the dump into a new branch</li> <li>✓ Fails listing every failed statement (instance kept)</li> <li>✓ Same for Postgres imports (I1)</li> <li>✓ Automated test</li> </ul>                                                                                                                     | Unit; VM §11         |
| <input checked="" type="checkbox"/> I6  | <b>From JSON / NDJSON</b> <ul style="list-style-type: none"> <li>✓ Columns inferred from top-level keys</li> <li>✓ An invalid document fails the import, naming its element or line</li> </ul>                                                                                                                                                                      | Unit; VM §10, §11    |
| <input checked="" type="checkbox"/> I7  | <b>From CSV</b> <ul style="list-style-type: none"> <li>✓ Table from the header, exact column names</li> <li><input type="checkbox"/> Typed columns (all text today)</li> </ul>                                                                                                                                                                                      | Manual               |
| <input checked="" type="checkbox"/> I8  | <b>Continuous replication &amp; cutover</b> <ul style="list-style-type: none"> <li>✓ Initial copy + streaming (logical replication)</li> <li>✓ Cutover keeps data (CLI)</li> <li>✓ Status and cutover from REST and the Import page</li> <li>✓ A replicating branch is never auto-suspended</li> <li>✓ Automated test (VM §5b) and driven in the browser</li> </ul> | Unit; VM §5b         |
| <input checked="" type="checkbox"/> I9  | <b>Guard handling during import</b> <ul style="list-style-type: none"> <li>✓ Guardrail paused, recording stays on</li> <li>✓ Re-armed even after a crash</li> </ul>                                                                                                                                                                                                 | Manual               |
| <input checked="" type="checkbox"/> I10 | <b>Live progress (CLI &amp; SSE)</b> <ul style="list-style-type: none"> <li>✓ Live log and progress bar</li> <li><input type="checkbox"/> psql subprocess output not streamed</li> </ul>                                                                                                                                                                            | Manual               |
| <input checked="" type="checkbox"/> I11 | <b>Import REST endpoints</b> <ul style="list-style-type: none"> <li>✓ From a connection string or an uploaded file</li> <li><input type="checkbox"/> Target name validation</li> <li>✓ Private, local and container sources for admins only; metadata refused</li> </ul>                                                                                            | Manual; unit; VM §1d |
| <input checked="" type="checkbox"/> I12 | <b>Web Import page</b> <ul style="list-style-type: none"> <li>✓ Database and file panels with live console</li> <li>✓ No longer claims a SQLite loader that doesn't exist</li> </ul>                                                                                                                                                                                | Manual               |
| <input checked="" type="checkbox"/> I13 | <b>Exact identifier fidelity</b> <ul style="list-style-type: none"> <li>✓ Case kept ( <code>userId</code> , <code>leaseAiChats</code> )</li> <li><input type="checkbox"/> Field order of JSON sources not kept</li> </ul>                                                                                                                                           | Unit; VM §10         |

## J ETL pipelines (experimental)

4 of 4 done

| ID                                     | Feature · sub-features                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Tested by   |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| <input checked="" type="checkbox"/> J1 | <b>Pipeline engine</b> <ul style="list-style-type: none"> <li>✓ Extract → raw schema → SQL models → tests, in a fresh branch</li> <li>✓ <code>ref()</code> / <code>source()</code> templating</li> <li><input type="checkbox"/> Dependency graph</li> <li><input type="checkbox"/> Incremental models</li> <li><input type="checkbox"/> Scheduling</li> <li>✓ A run replaces only a branch the pipeline made</li> <li>✓ API sources are connection strings, never server paths</li> </ul> | Unit; VM §9 |
| <input checked="" type="checkbox"/> J2 | <b>Data-quality tests</b> <ul style="list-style-type: none"> <li>✓ <code>not_null</code>, <code>unique</code>, <code>accepted_values</code>, <code>row_count_min</code>, custom</li> <li>✓ Failure → exit 1, data kept</li> </ul>                                                                                                                                                                                                                                                         | Unit; VM §9 |

|   |           |                                                                                                                |                                |        |
|---|-----------|----------------------------------------------------------------------------------------------------------------|--------------------------------|--------|
| ✓ | <b>J3</b> | <b>Stored pipelines &amp; run history</b><br>✓ Saved per user, run over the API, history<br>□ Read a run's log | □ Mark runs whose process died | Manual |
| ✓ | <b>J4</b> | <b>Pipeline editor (web)</b><br>✓ Create, edit, run, results, history                                          | □ Confirm before delete        | Manual |

## K APIs, SDKs & agents 6 of 6 done

| ID          | Feature · sub-features                                                                                                                                                                                                                   | Tested by                                                                                                                                                                                   |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ✓ <b>K1</b> | <b>Control-plane REST API</b><br>✓ Branches, query, Blackbox, policy, impact, import, pipelines, keys<br>□ Versioned paths ( <code>/api/v1</code> )                                                                                      | Unit; VM §1; v2 §0–§8; VM §1d<br>✓ Exact-origin CORS, branch-name check, body limits, timeouts<br>✓ Branches owned by their maker; others 404 (owner, admin, or main)                       |
| ✓ <b>K2</b> | <b>OpenAPI spec</b><br>✓ Served, Blackbox 2.0 endpoints documented                                                                                                                                                                       | Unit (drift check)<br>✓ Matches the code (import fields, status codes, auth routes, <code>/blackbox</code> aliases)<br>✓ Drift check in CI                                                  |
| ✓ <b>K3</b> | <b>Python SDK</b><br>✓ Branches, query, Blackbox<br>□ Import, pipelines, keys, agents                                                                                                                                                    | Manual<br>□ Tests<br>□ Publish to PyPI                                                                                                                                                      |
| ✓ <b>K4</b> | <b>TypeScript SDK</b><br>✓ Same client for Node and browsers<br>□ Tests                                                                                                                                                                  | Manual<br>□ Publish to npm                                                                                                                                                                  |
| ✓ <b>K5</b> | <b>Agent Branch API</b><br>✓ A branch per agent, own non-superuser role<br>✓ Provenance (task, parent, session)<br>✓ DSN reachable from the host (gateway-routed, branch-scoped key)                                                     | Unit; VM §5; v2 §1c, §7<br>✓ Cap, listing and TTL count suspended branches<br>✓ Superuser switch only in <code>-tags insecure</code> builds<br>✓ Agents and their keys belong to the caller |
| ✓ <b>K6</b> | <b>MCP server</b><br>✓ Branches, <code>run_sql</code> , Blackbox tools<br>✓ <code>policy_check</code> , <code>execute_change</code> , impact, diff<br>✓ Requires an API key and acts as that account; a scoped key is held to its branch | Unit; v2 §1c–§8<br>✓ Client config & 16-tool reference ( <code>docs/mcp.md</code> )<br>✓ Same ownership rule as the REST API                                                                |

## L Web console 11 of 11 done

| ID          | Feature · sub-features                                                                                                                                                                                                          | Tested by                                                                                                                                                                       |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ✓ <b>L1</b> | <b>Embedded single-origin UI</b><br>✓ Served by the engine at <code>:8080</code>                                                                                                                                                | Manual<br>□ Automated UI tests (whole area)                                                                                                                                     |
| ✓ <b>L2</b> | <b>Login &amp; signup</b><br>✓ Log in / sign up tabs, OAuth buttons                                                                                                                                                             | Manual                                                                                                                                                                          |
| ✓ <b>L3</b> | <b>Dashboard</b><br>✓ Live tiles                                                                                                                                                                                                | Manual<br>✓ Branch table: create, suspend, resume, delete, copy DSN                                                                                                             |
| ✓ <b>L4</b> | <b>SQL console &amp; browser</b><br>✓ Schema sidebar, rows, structure, indexes<br>✓ Values shown as Postgres writes them ( <code>uuid</code> , <code>interval</code> , <code>inet</code> , <code>bytea</code> )<br>✓ SQL editor | Manual; API in v2 §6<br>✓ System tables hidden by default, with a show button<br>✓ Runs as the signed-in user; one-run override switch<br>✓ Per-rule override for policy blocks |
| ✓ <b>L5</b> | <b>Blackbox page</b><br>✓ Entries per branch with badges<br>✓ Table / risk / date filters, and "Clear filters"                                                                                                                  | v2 §5b; browser<br>✓ Verify button; a row expands to its statement and session                                                                                                  |

|   |                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                         |                                                         |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| ✓ | <b>L6 API keys page</b><br>✓ Create, copy, list, revoke                                                                                                                                                                                                                                                     | ✓ A new key is shown inside its connection string (E3's next step)                                                                                                                                      | Manual; browser review 21 Sep                           |
| ✓ | <b>L7 Landing, Guide &amp; Docs pages</b><br>✓ Landing, developer guide, reference                                                                                                                                                                                                                          | ✓ Refreshed against the code (https, sslmode, MCP, Blackbox, import routes)                                                                                                                             | Type-check + build                                      |
| ✓ | <b>L8 Blackbox integrity page</b><br>✓ Verify, checkpoint, export, branch-before<br>✓ Used signed in; every result matched the CLI                                                                                                                                                                          | ○ Entry list needs a refresh to show a new change                                                                                                                                                       | Type-check + build; browser review 16 Sep               |
| ✓ | <b>L9 Policies page</b><br>✓ Rules, warn/block, custom rules, preview, evaluations<br>✓ Used signed in; matched the CLI, wrote only to the chosen branch                                                                                                                                                    | ✓ Non-admin change refused with a readable message<br>✓ Who can override: list, grant, revoke                                                                                                           | Type-check + build; API in v2 §6; browser review 16 Sep |
| ✓ | <b>L10 App sidebar layout</b><br>✓ Every link opens its page, highlighted, with breadcrumb<br>✓ Collapse persists across a reload; theme toggle persists                                                                                                                                                    | ✓ No console errors on any page<br>□ Under-860px drawer (needs a manual look)                                                                                                                           | Type-check + build; browser review 16 Sep               |
| ✓ | <b>L11 Design system &amp; FoxByte theme</b><br>✓ Tokens for colour, type, space, radius, elevation and motion; components use no raw values<br>✓ Ember palette in both themes; filled controls legible ( <code>--accent-ink</code> )<br>✓ Fox mark from theme variables, wordmark from brand.json, favicon | ✓ Console, Blackbox, Policies and API keys re-laid out; one table style<br>✓ Every page reviewed in both themes; console checked at 375px<br>□ Not yet reviewed against a running engine with real data | Type-check + build; browser review 21 Sep               |

## M Build, CI & releases

7 of 7 done

| ID   | Feature · sub-features                                                                                                                                                                                                                 | Tested by                                                                                                                                                                                                                                                                                                                                                |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ✓ M1 | <b>Makefile targets</b> <ul style="list-style-type: none"><li>✓ build, vet, test, integration suites</li><li>✓ Suites run only in a throwaway test VM (guarded)</li></ul>                                                              | CI                                                                                                                                                                                                                                                                                                                                                       |
| ✓ M2 | <b>Linux CI</b> <ul style="list-style-type: none"><li>✓ Build, vet, test, gofmt</li><li>✓ Release script tests, checklist sync check<ul style="list-style-type: none"><li>□ VM integration suites in CI</li></ul></li></ul>            | CI                                                                                                                                                                                                                                                                                                                                                       |
| ✓ M3 | <b>Windows CI</b> <ul style="list-style-type: none"><li>✓ Build, vet, test, Pester, installer encoding</li><li>✓ Guard tests read backslash paths, and generated files are pinned to LF (21 Sep)</li></ul>                             | CI <ul style="list-style-type: none"><li>□ End-to-end WSL install test</li></ul>                                                                                                                                                                                                                                                                         |
| ✓ M4 | <b>Releases &amp; GHCR image</b> <ul style="list-style-type: none"><li>✓ Binaries for 5 platforms + fox-verify</li><li>✓ SHA256SUMS merged safely between jobs</li></ul>                                                               | Script tests in CI <ul style="list-style-type: none"><li>✓ Fails unless fox update 's files are published</li><li>✓ Multi-arch Postgres image on GHCR</li></ul>                                                                                                                                                                                          |
| ✓ M5 | <b>WSL distro &amp; ZFS builds</b> <ul style="list-style-type: none"><li>✓ Prebuilt distro image</li><li>✓ Downloaded and verified by the installer (seen on Windows)</li><li>✓ Preloaded image tagged as the engine runs it</li></ul> | Actions artifacts; Pester; Windows run 16 Sep <ul style="list-style-type: none"><li>✓ Imported on a fresh Windows install; preloaded images loaded</li><li>✓ Published zstd-compressed, with the gzip size measured beside it in every build log</li><li>🟡 zstd unpack on a real Windows install (unit-tested only); the gzip image was 685 MB</li></ul> |

|                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                             |
|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
|  | <div data-bbox="159 103 399 145"> <b>M6 Postgres image</b> </div> <ul style="list-style-type: none"> <li>✓ PostgreSQL 18 + wal-g for a fresh install; one image per supported major (16, 18)</li> <li>✓ The image follows the data: an install on 16 keeps running 16 after an update</li> <li>✓ A branch on the wrong major is refused before <code>docker run</code>, with the way out</li> </ul> <ul style="list-style-type: none"> <li>✓ The major lives in one place, and a test holds the Dockerfile, release matrix and Windows preload to it</li> <li>✓ All three suites pass on PostgreSQL 18.6: 108 + 277 + 51 checks (21 Sep)</li> <li>✓ A command that moves an install from 16 to 18 (D6), and a portable backup to take first (D5)</li> </ul> <div data-bbox="686 369 798 392"> <input type="checkbox"/> pgvector         </div> | Unit; all three suites on 18                                |
|  | <div data-bbox="159 403 478 448"> <b>M7 Product name in one file</b> </div> <ul style="list-style-type: none"> <li>✓ <code>brand.json</code> → Go, shell, web and installer blocks, regenerated by <code>make brand</code></li> <li>✓ Stored names brand-free and frozen: a rename never migrates an install</li> <li>✓ Retired env prefixes, state directory, VM names and installs still handled</li> </ul> <ul style="list-style-type: none"> <li>✓ Guards: retired names anywhere, product names in stored SQL, durable names spelt twice</li> <li>🟡 The sweep is text substitution: review its diff (it over-reached twice)</li> </ul>                                                                                                                                                                                                    | Unit (brand, gen, oldnames), in CI via <code>go test</code> |

## N Not started (roadmap)

11 items

Planned or discussed, nothing built yet. When work starts, move the row into its area with the next id there.

| ID                         | Item · scope                                                                                                                                         | Related |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| <input type="checkbox"/> - | <b>Multi-tenant projects &amp; roles</b><br><input type="checkbox"/> Projects / organisations <input type="checkbox"/> Per-branch permissions (RBAC) | G1, H6  |
| <input type="checkbox"/> - | <b>Branch merge</b><br><input type="checkbox"/> Apply one branch's schema changes to another                                                         | H18     |
| <input type="checkbox"/> - | <b>Scheduled backups &amp; retention</b>                                                                                                             | D2      |
| <input type="checkbox"/> - | <b>Automatic HA failover</b><br><input type="checkbox"/> Health detection, fencing <input type="checkbox"/> Multi-host                               | D4      |
| <input type="checkbox"/> - | <b>Pipelines: scheduling, incremental / CDC extract, DAG view</b>                                                                                    | J1      |
| <input type="checkbox"/> - | <b>More import sources: Cassandra, DynamoDB</b>                                                                                                      | I1–I8   |
| <input type="checkbox"/> - | <b>pgvector in the Postgres image</b>                                                                                                                | M6      |
| <input type="checkbox"/> - | <b>Cloud write-once anchors</b> (S3 Object Lock etc.)                                                                                                | H10     |
| <input type="checkbox"/> - | <b>Release signatures</b>                                                                                                                            | A9, M4  |
| <input type="checkbox"/> - | <b>Hosted cloud tier</b>                                                                                                                             | —       |
| <input type="checkbox"/> - | <b>Networked storage disaggregation</b> (README roadmap; see the storage engine note on page 1)                                                      | C1, D1  |

### Out of scope by decision

A custom storage engine (PostgreSQL's own engine is used, for native transaction speed), sharding, Raft / distributed transactions, multi-region, HTAP.